TIMBER ATLAS

Quick User Guide

Open the 33-member viewer

01 Open the model

Open the shared link and wait for loading. This public viewer contains a fixed snapshot of 33 timber members. No Rhino, Grasshopper, GitHub account, or AI is required.

02 Explore and select

Drag to rotate, scroll to zoom, and right-drag to pan. Click a member to highlight it and inspect dimensions, joints, and contacts. Selection preserves the camera. Use the overview control to frame the model.

03 Six-face drawings

Select a member and open its six-face drawings. Choose one face or all six on one sheet. Scroll to zoom, drag to pan, and click a cut for details. SVG export and Print / Save PDF are available. Press Esc to return to 3D.

04 Measure distance

Enable measurement in the main 3D toolbar. Snap to real straight-edge endpoints or midpoints, then click two points for distance in mm. Clear or Esc clears the result; a third point starts again. Layout and joint-diagnostic views do not support measurement.

05 Assembly steps

Open Assembly. Use Previous, Next, Play, or Pause, and adjust insertion progress to inspect the current member and direction. Obstruction diagnostics identify a candidate direction and blocking member. Esc returns to 3D. The browser plays the saved plan; it does not recalculate collisions for an arbitrary reordered sequence.

06 Interface language

Choose English, Japanese, or Chinese at the top right. Member IDs and model data are unchanged. Viewing, measurement, and assembly playback do not modify the original model.

Scope and limitations

Checks cover the implemented fixed-orientation, straight-line insertion candidates. They do not certify structural safety or construction feasibility, and exclude temporary supports, tools, lifting equipment, and personnel clearance. Verify dimensions, allowances, and fabrication methods before manufacture. The expanded 24-case validation remains deferred; this is a review draft.

Requirements and installation

Two distinct workflows

A. View the published model

Use an internet-connected browser with WebGL enabled (Edge tested). No Rhino, Grasshopper, Python, Codex, AI, or GitHub account is required. The public site contains one saved 33-member model and a precomputed plan; it does not analyse new geometry.

B. Analyse your own model

The current setup targets Windows and requires a working Rhino 8 installation with Grasshopper plus the supplied Timber Atlas plugin. Python 3 and a browser are needed for the local web viewer, not for GH geometry calculations. Python server code uses the standard library; no pip packages are required. Git is optional for ZIP users. Prebuilt plugins are included: Visual Studio and the .NET SDK are not needed to use them.

1. Download the complete repository

Sign in to GitHub with an account authorized for likairesearch-jpg/Timber-Atlas. Download and extract the repository ZIP, or clone it. Keep build/, tools/, viewer/, and examples/ together in a writable folder. Do not install only a .gh or .gha file. The Live component locates viewer/ relative to build/v0.4.10.0/.

2. Register the plugin

Close Rhino. In PowerShell, change to the extracted repository root and run:

```
.\tools\Install-Plugin.ps1
```

The script writes a Grasshopper .ghlink to this folder. Keep the folder in place. Reopen Rhino and Grasshopper, then open examples/Current_Measurement_v010.gh. Confirm components 01-13 load before editing your model.

3. Start the local viewer first

Install Python 3 and make python available on PATH. Open a new PowerShell in the repository root:

```
python --version
.\tools\Start-Viewer.ps1
```

Important: GH's automatic server launch currently looks for a Codex-bundled Python. On a new PC without that runtime, start the script above first; it falls back to Python on PATH. Codex is not required. If necessary, run python viewer/serve.py --port 8767 in a terminal and leave it running.

4. Connect your analysis

Use 01 Timber Group for your axes and sections in mm; Merge > 03 Detect Joints > 04 Cut Joints > 13 Assembly Analysis > 10 Web Viewer / 09 Export Assembly. Supply the appropriate rules to 04. Enable Run for analysis, Live for synchronization, and press Open. Use the session URL from 10, not just the server root, for live data. Change 09 Write from False to True to export JSON.

Moving to another computer

Recovery, checks, and known limits

Files to transfer separately

The repository contains the example GH file and the published model JSON. Your working Rhino .3dm files, external referenced geometry, unsaved GH changes, and private files outside the repository are not automatically transferred. Save and copy these separately. Reassign Rhino-referenced axes if necessary. Set Export Assembly Path to a writable path on the new PC; old C: paths will not be valid.

Common setup issues

If Windows blocks downloaded files, unblock only this trusted repository's downloaded archive through its Properties before extraction; do not disable system-wide protections. If PowerShell blocks a script, follow the computer's policy or administrator guidance. If Install-Plugin reports another .ghlink, inspect the existing target and resolve it before registering again; do not delete other plugins. Missing component errors require checking plugin registration and restarting Rhino.

Ports and Python

If python --version opens the Microsoft Store or fails, repair the Python installation/PATH and reopen the terminal. If port 8767 is occupied, run:

```
.\tools\Start-Viewer.ps1 -Port 8768
```

Set the GH Web Viewer Port to the same value. Live=False pauses data; Open alone does not publish a new model. A loopback URL works only on the same computer.

Installation acceptance checklist

1. No missing Timber Atlas components in the example.
2. Inspect component Info for errors after recalculation.
3. Start the viewer, enable Live, then Open; confirm the expected model count.
4. Change a parameter in a copy of the example and confirm synchronization.
5. Test selection, measurement, drawings, and assembly playback when a plan is present.
6. Export JSON to the new location, import it into the full local viewer, and confirm it loads. The public snapshot site intentionally has no import control.

View a downloaded snapshot without GH

Extract the share-package ZIP. In its folder run python run_local.py, then open http://127.0.0.1:8780/. Keep the server running. This needs only Python 3 and a browser. Double-clicking index.html is not the supported startup method. Send the public hosted link for remote viewing.

Continue development with a new assistant

Open the complete repository as the workspace. Read README.md, docs/planning/PROJECT_CHARTER_AND_ROADMAP.md, PROGRESS.md and AGENT_COLLABORATION.md in docs/planning/, then the latest QA records. Inspect git status before editing. Conversation history, machine credentials, and local Rhino connections do not travel with a clone. Reconnect tools separately only if needed for development.

Verification boundary

Documentation has been checked against tracked installer/server code and a fresh repository extraction. A full clean-PC Rhino installation has not been tested. macOS installation and a minimum Python compatibility version are not established. Viewer-only users need none of the development setup above.